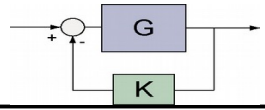


ME 2801

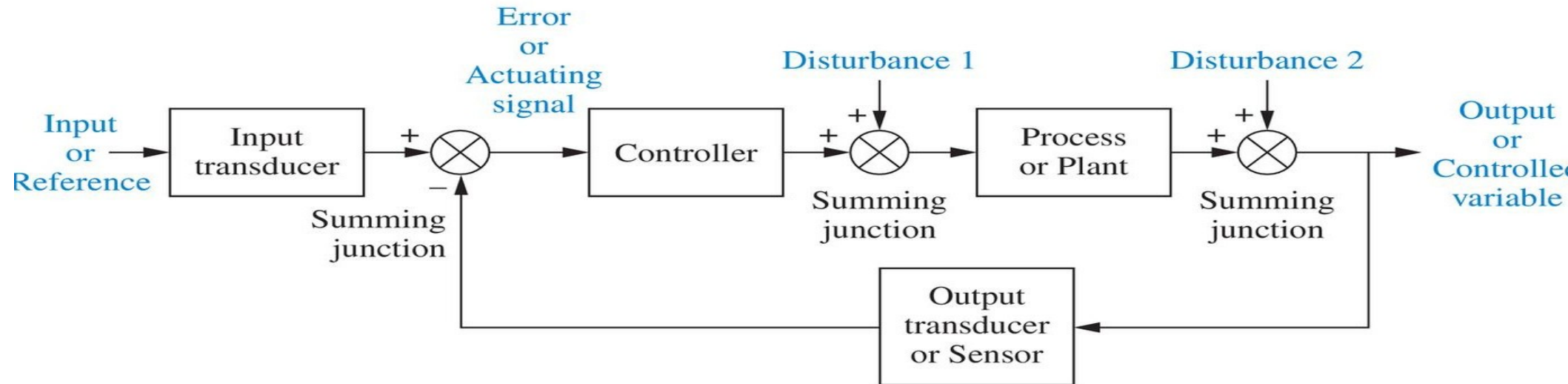
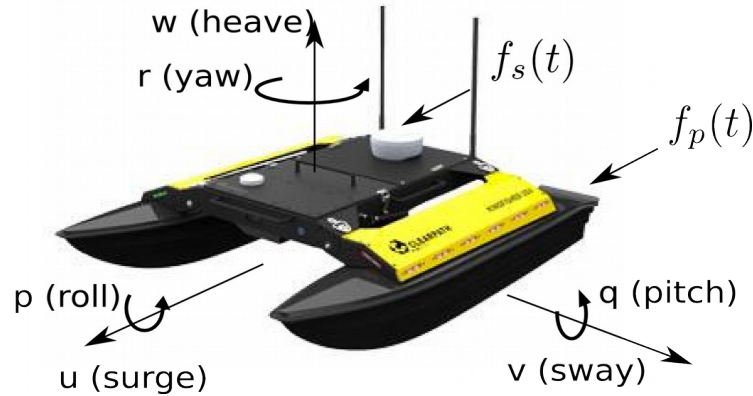
Introduction to Control Systems

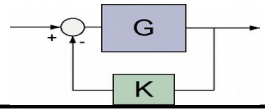
W2: Modeling
Mechanical, 1 DOF

Closed-Loop System



Fill in the blanks for USV heading/speed example...



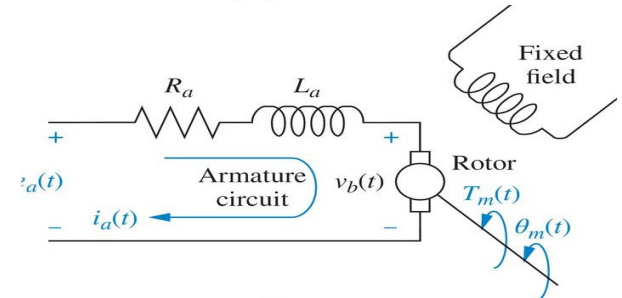
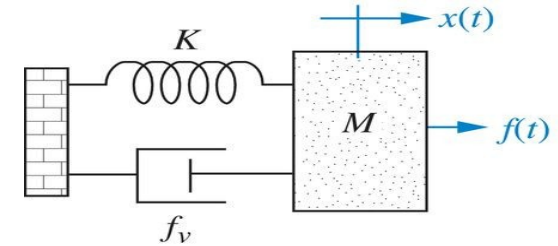
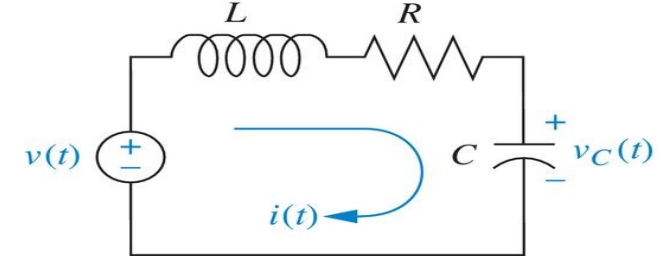
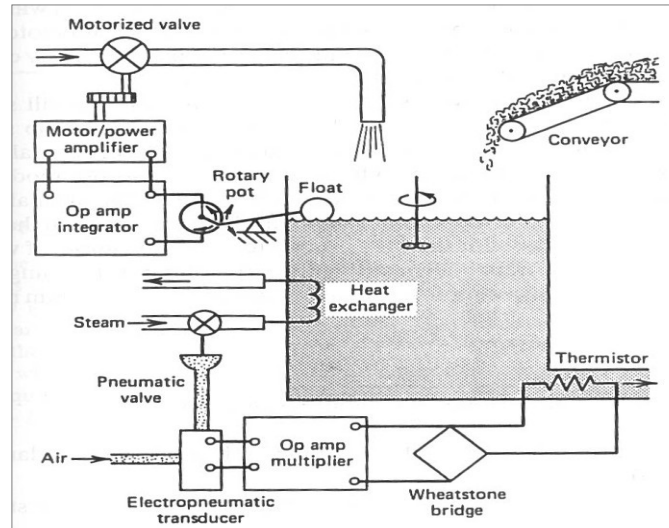
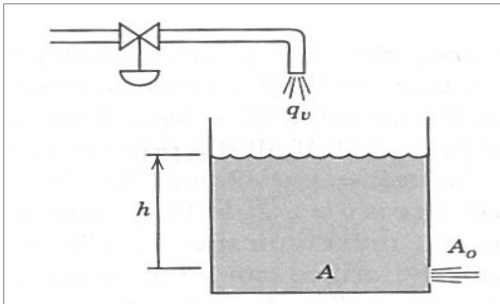


“Lumped Parameter Models”

- Discrete element representation
- ODEs not PDEs

Examples

- Mechanical: translational/rotational
- Electrical circuits
- Thermal
- Fluid



Lumped Parameter Elements

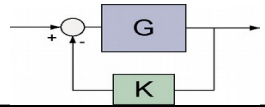
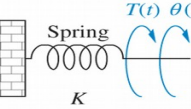
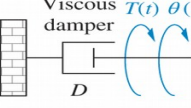
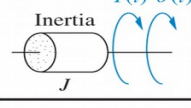
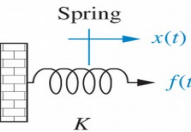
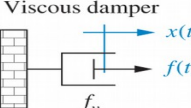
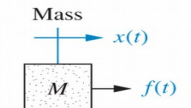


TABLE 2.5 Torque-angular velocity, torque-angular displacement, and impedance rotational relationships for springs, viscous dampers, and inertia

Component	Torque-angular velocity	Torque-angular displacement	Impedance $Z_M(s) = T(s)/\theta(s)$
 Spring K	$T(t) = K \int_0^t \omega(\tau) d\tau$	$T(t) = K\theta(t)$	K
 Viscous damper D	$T(t) = D\omega(t)$	$T(t) = D \frac{d\theta(t)}{dt}$	Ds
 Inertia J	$T(t) = J \frac{d\omega(t)}{dt}$	$T(t) = J \frac{d^2\theta(t)}{dt^2}$	$J s^2$

Note: The following set of symbols and units is used throughout this book: $T(t)$ – N-m (newton-meters), $\theta(t)$ – rad (radians), $\omega(t)$ – rad/s (radians/second), K – N-m/rad (newton-meters/radian), D – N-m-s/rad (newton-meters-seconds/radian), J – kg-m² (kilograms-meters² – newton-meters-seconds²/radian).

TABLE 2.4 Force-velocity, force-displacement, and impedance translational relationships for springs, viscous dampers, and mass

Component	Force-velocity	Force-displacement	Impedance $Z_M(s) = F(s)/x(s)$
 Spring K	$f(t) = K \int_0^t v(\tau) d\tau$	$f(t) = Kx(t)$	K
 Viscous damper f_v	$f(t) = f_v v(t)$	$f(t) = f_v \frac{dx(t)}{dt}$	$f_v s$
 Mass M	$f(t) = M \frac{dv(t)}{dt}$	$f(t) = M \frac{d^2x(t)}{dt^2}$	$M s^2$

Note: The following set of symbols and units is used throughout this book: $f(t)$ = N (newtons), $x(t)$ = m (meters), $v(t)$ = m/s (meters/second), K = N/m (newtons/meter), f_v = N-s/m (newton-seconds/meter), M = kg (kilogram newton-seconds²/meter).

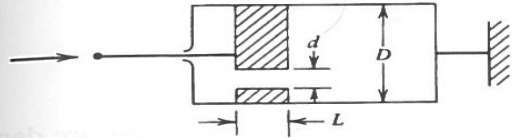
TABLE 2.3 Voltage-current, voltage-charge, and impedance relationships for capacitors, resistors, and inductors

Component	Voltage-current	Current-voltage	Voltage-charge	Impedance $Z(s) = V(s)/I(s)$	Admittance $Y(s) = I(s)/V(s)$
 Capacitor	$v(t) = \frac{1}{C} \int_0^t i(\tau) d\tau$	$i(t) = C \frac{dv(t)}{dt}$	$v(t) = \frac{1}{C} q(t)$	$\frac{1}{Cs}$	Cs
 Resistor	$v(t) = Ri(t)$	$i(t) = \frac{1}{R} v(t)$	$v(t) = R \frac{dq(t)}{dt}$	R	$\frac{1}{R} = G$
 Inductor	$v(t) = L \frac{di(t)}{dt}$	$i(t) = \frac{1}{L} \int_0^t v(\tau) d\tau$	$v(t) = L \frac{d^2q(t)}{dt^2}$	Ls	$\frac{1}{Ls}$

Note: The following set of symbols and units is used throughout this book: $v(t)$ – V (volts), $i(t)$ – A (amps), $q(t)$ – Q (coulombs), C – F (farads), R – Ω (ohms), G – Ω (mhos), L – H (henries).

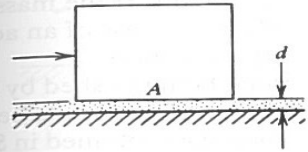
Table 4.2-1 Damping Constants of Common Elements

Dashpot (laminar flow)



$$c = 8\pi\mu L \left[\left(\frac{D}{d} \right)^2 - 1 \right]^2$$

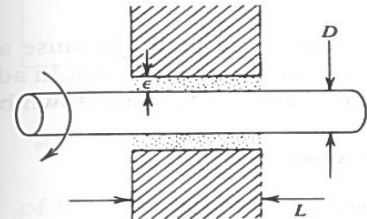
Sliding viscous friction



$$c = \frac{A\mu}{d}$$

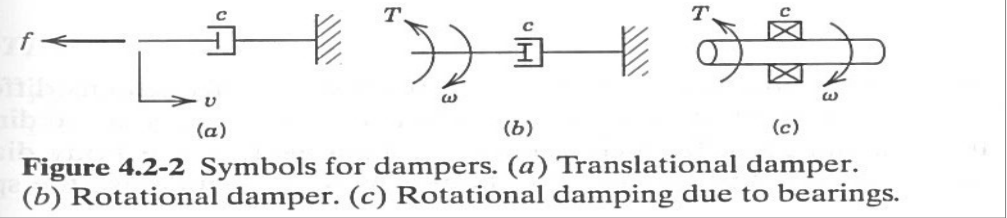
A = lubricated area
 d = film thickness

Journal bearing

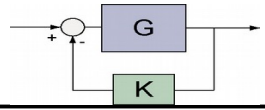


$$c = \frac{\pi D^3 L \mu}{4\epsilon}$$

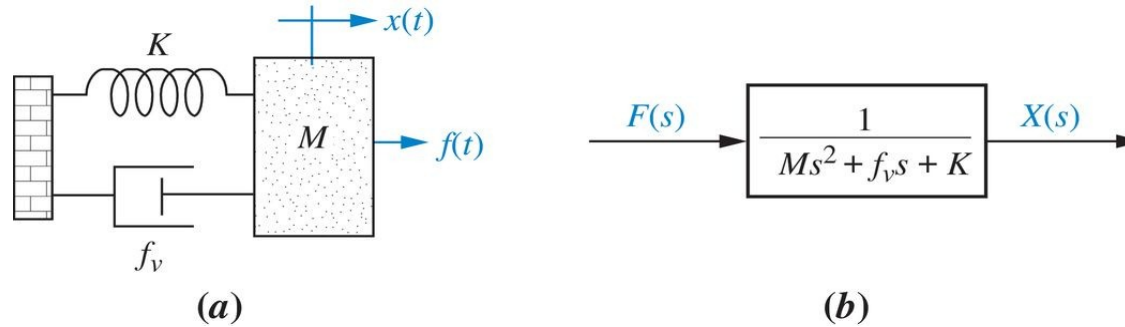
ϵ = radial clearance



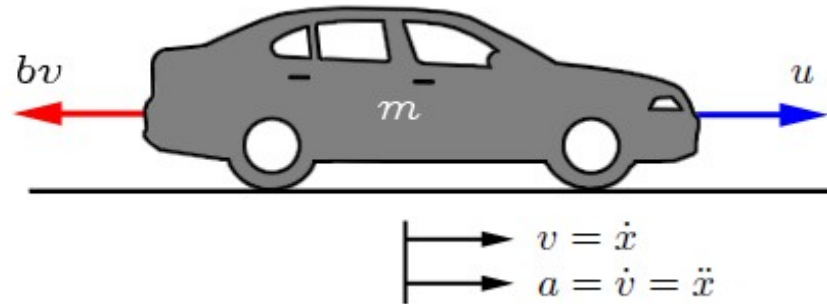
Examples: 1 DOF Translation



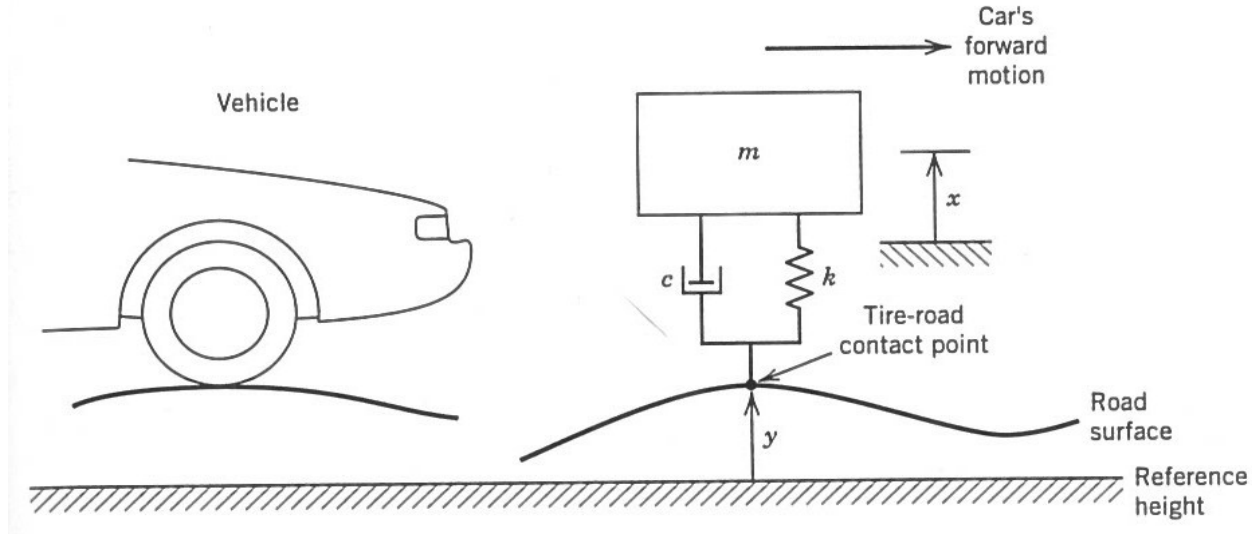
Mass, Spring, Damper



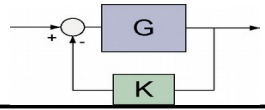
Vehicle Model



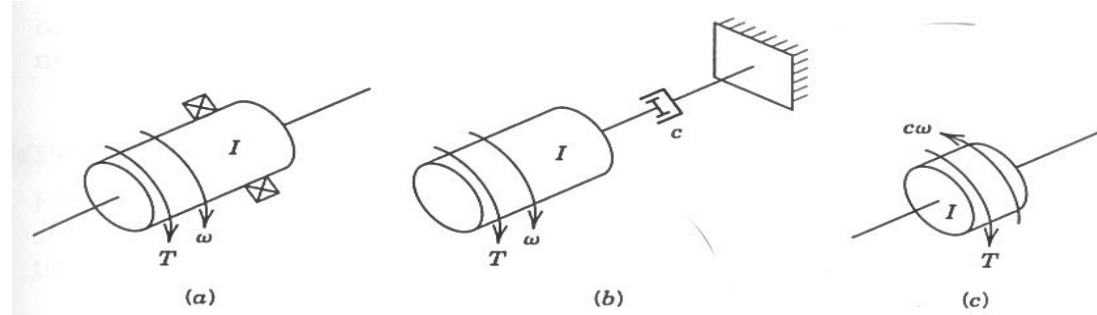
Quarter Car Model



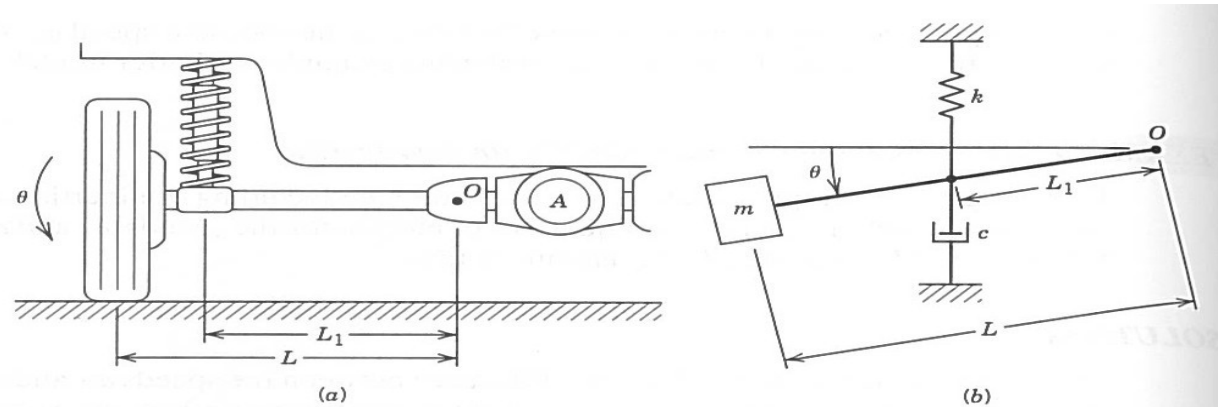
Examples: 1 DOF Rotation



Axle and journal bearing

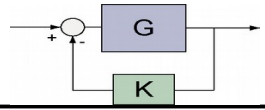


Suspension Swing Arm





HW2 Preview



Inverse Laplace w/ Complex Roots

Transfer Functions

- Type 1 Problems: Given an ODE, find the transfer function

Transfer Function Representation w/ MATLAB (x2)

- Using `zpk()`, `tf()` and `roots()`

Mechanical Modeling – 1 DOF

- Translational System Model
- Rotational System Model

USV Model

- Two independent DOF's – surge and yaw